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To cite this article: Ren Zhao-Xian, Chen Deng-Kai, Gong Yu-Han, Niu Shi-Fan, An Wei-Lan, Liu Zhuo & Wei Su-Yao (18 Aug 2025): Exploring the complex mapping between user perception and product form: a case study of air purifiers, Journal of Engineering Design, DOI: 10.1080/09544828.2025.2546834

To link to this article: <https://doi.org/10.1080/09544828.2025.2546834>



Published online: 18 Aug 2025.



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Exploring the complex mapping between user perception and product form: a case study of air purifiers

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ABSTRACT

This study integrates Kansei Engineering with a Genetic Algorithm (GA)-optimized Backpropagation (BP) neural network to enhance the emotional appeal of air purifier design. Ten affective descriptors related to the appearance of air purifiers were first identified. Their subjective and objective weights were calculated using the Analytic Hierarchy Process and Coefficient of Variation methods, respectively. The most representative descriptors were then selected through the coupling coordination degree model. Using morphological analysis, the product's external components were systematically deconstructed and encoded to generate thirty distinct design combinations. Emotional response values for each design were quantitatively evaluated based on the selected descriptors. A GA-optimized BP neural network was employed to model the complex mapping between form parameters and emotional responses. The trained model was subsequently applied to predict emotional evaluations and determine the optimal design configuration. The final design was validated through empirical user testing. Results indicate that the proposed approach can effectively facilitate the identification and prediction of form-emotion relationships, offering a promising methodological framework for emotionally driven product form development.

ARTICLE HISTORY

Received 30 October 2023
Accepted 5 August 2025

KEYWORDS

Kansei engineering; genetic algorithm; backpropagation neural network; product form design; air purifier

1. Introduction

In the context of increasingly intense market competition, product appearance design has emerged as a pivotal factor influencing consumer purchasing decisions (Blijlevens, Creusen, and Schoormans 2009), particularly for home appliances such as air purifiers. Beyond core functionality, the visual form of these products exerts a substantial influence on the overall user experience (Shaw and Nickpour 2024; Wang, Cheng, and Liang 2021). As consumers

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place growing emphasis on emotional engagement, conventional design approaches often fail to adequately address the escalating demand for personalisation and emotional resonance. Consequently, scientifically optimising product appearance to more effectively embody emotional value has emerged as a central challenge in product design.

Kansei Engineering (KE), a methodology that transforms user emotions into product design parameters, has been widely applied to optimise product appearance, colour, and material choices (Yang, Liu, and Ye 2023). To improve its systematicity and scientific rigour, researchers have integrated KE with complementary approaches such as the Fuzzy Analytic Hierarchy Process and Quality Function Deployment (QFD) (Fu et al. 2024). To enhance the reliability and consistency of emotional data interpretation, some studies have incorporated tools such as eye-tracking (Yang et al. 2021) and electrophysiological signal analysis (Ding et al. 2023) to extract user perceptions and refine design solutions. While these approaches have advanced the precision of affective preference identification, they continue to rely heavily on subjective judgment and exhibit limitations in modelling complex, nonlinear relationships between user perception and product form.

With the advancement of intelligent algorithms, there has been growing interest in applying machine learning techniques to explore the intricate connections between emotional responses and design parameters. For instance, (Zhu et al. 2024) developed an optimisation model leveraging Backpropagation (BP) neural networks to examine how Apple product design features relate to user satisfaction, while (Jing et al. 2023) combined BP neural networks with EEG data to infer users' latent psychological preferences. Although BP networks are proficient in learning nonlinear relationships, they are constrained by issues such as high computational complexity and susceptibility to local minima, which limit their global optimisation capabilities. To address these limitations, Genetic Algorithms (GA) have been utilised to optimise the initial weights and structural parameters of BP networks, thereby improving training efficiency and prediction accuracy.

Motivated by these challenges, this study proposes an emotional optimisation framework that integrates hybrid KE with intelligent algorithms to support data-driven product appearance design. The primary objective is to more effectively extract and quantify users' affective needs while modelling the complex, nonlinear mapping between emotional perception and product form in a scientifically robust manner. Specifically, this research addresses the limitations of traditional approaches in handling multi-label, nonlinear emotional expression. The key contributions of this study are as follows:

- A fusion weighting method combining the Analytic Hierarchy Process (AHP) and Coefficient of Variation (CV) is developed to calculate both subjective and objective weights of emotional descriptors. The Coupling Coordination Degree (CCD) model is then introduced to improve the rationality and consistency of weight allocation, thereby reducing reliance on expert judgment and identifying the most representative emotional vocabulary.
- Morphological analysis is employed to modularise and encode product form components, generating quantifiable and structured parameter combinations that serve as inputs for neural network models.
- A BP neural network is constructed to model the nonlinear relationship between product form parameters and emotional descriptors. To improve model convergence and

predictive performance, a GA is applied to optimise the network's initial weights and structural configuration. This integration enhances the generalisation and global search capabilities of the model in emotional classification tasks, enabling the identification of optimal product designs tailored to specific affective needs.

The remainder of this paper is organised as follows: Section 2 reviews related literature. Section 3 details the proposed methodology. Section 4 presents a case study on air purifier design. Section 5 discusses practical applications and validation. Section 6 concludes the study and outlines directions for future research.

2. Literature review

2.1. *Methods for screening consumer emotional needs*

Accurately identifying and selecting consumer emotional needs is a core issue in KE, aiming to transform complex and dynamic emotional information into actionable product design parameters. In recent years, numerous innovative methods have been developed to support the screening of emotional needs, offering both theoretical frameworks and practical tools for product design.

Among these methods, the fuzzy Delphi method and network analysis are widely used quantitative approaches. For instance, (Yao et al. 2022) utilised the fuzzy Delphi method to select evaluation criteria and applied network analysis to determine the weights of design factors. This approach provides a systematic framework for balancing subjective expert judgment with objective data, facilitating the integration of multidimensional information. Semantic analysis and satisfaction modelling also play important roles in emotional needs research. (Ye, He, and Liu 2024) investigated wearable smart products by applying semantic analysis, the KANO model, and satisfaction analysis to classify consumer needs. Based on the influence of consumer behaviour, they proposed targeted design strategies, offering high practical value for need stratification and prioritisation. In the context of sustainable design, QFD has been frequently used to decompose and prioritise emotional needs. (Vinodh et al. 2017) implemented a fuzzy QFD method combined with House of Quality techniques to identify key sustainable parameters for consumer electronics and determine design tool priorities, thereby offering solid technical support for sustainability-oriented design. Artificial intelligence has further expanded the methodological boundaries in this area. (Fan et al. 2021) proposed a product design approach based on a multidimensional evaluation system that combines genetic algorithms with neural networks. This system effectively integrates consumer aesthetic preferences into the design process, enabling precise emotional mapping for optimisation. Moreover, VR is becoming an increasingly important tool for emotional needs research (Liu and Yang 2022), using a hairdryer as a case study, integrated VR with KE and developed a comprehensive experimental protocol, from affective vocabulary collection to four-dimensional morphological analysis. Their findings show that VR significantly enhances the accuracy and reliability of emotional preference predictions, providing a novel technological pathway for experimental design research. In summary, these diverse methods, ranging from quantitative techniques to AI and VR applications, collectively form a multilayered framework for emotional need screening, each offering distinct advantages.

However, several limitations persist. Most notably, there is a lack of integrated approaches that simultaneously incorporate both subjective and objective factors, especially when performing in-depth quantitative analyses of emotional features. Methods like fuzzy Delphi and QFD often depend heavily on expert input, which can be time-consuming and suffer from limited reproducibility. Additionally, traditional statistical or rule-based techniques are often inadequate in handling multivariate interactions and nonlinear relationships, rendering them less effective in managing high-dimensional emotional data in complex design scenarios.

To overcome these limitations, the present study integrates the AHP and CV to calculate both subjective and objective weights. Furthermore, the CCD method is introduced to identify the most representative affective features. This ensures a comprehensive and accurate analysis of emotional needs. Based on morphological analysis, product appearance features are deconstructed and encoded to automatically generate multiple design schemes. Emotional evaluations are then conducted through expert assessments, allowing for rapid data collection while mitigating the inefficiencies of traditional approaches.

2.2. Application of intelligent algorithms in product design

With the rapid development of artificial intelligence technologies, the application of intelligent algorithms in product design has advanced significantly, becoming a critical tool for shape optimisation, personalised design, and precision enhancement.

The integration of GA with deep learning has emerged as a focal point in design optimisation. For example, (Behúnová, Zemanová, and Behún 2023) proposed a method that combines GA with convolutional neural networks to improve design efficiency, performance, and innovation. This integration enables more accurate modelling of design objectives and accelerates the optimisation process. Enhancements in fitness allocation strategies have also introduced new possibilities for personalised design. (Xue 2019) introduced an interactive GA incorporating traditional design theory with a composite fitness allocation strategy. Through hierarchical optimisation, this method effectively addresses the personalised and diversified needs of users, enhancing the alignment between user demands and design parameters, and providing theoretical support for intelligent product development. In the field of emotion-driven intelligent design, (Cong et al. 2023) developed a smart product-service system based on self-organising maps. By analysing users' emotional feedback toward traditional products, this approach refined the initial design scheme for a smart electric bicycle service system. The results demonstrated that integrating emotional drivers with intelligent algorithms significantly improved user satisfaction. Researchers have also explored intelligent task and team optimisation. (Sun et al. 2024) proposed a task optimisation method based on implementation intention matching. This method constructs team models, assesses intention generalisation, computes recommendation values, and introduces a time loss model, applying Pareto optimisation to achieve more accurate team-task matching. Case studies in smart cabin environments confirmed improvements in task allocation accuracy and design efficiency. In the context of data-driven customisation, (Powell et al. 2024) proposed an intelligent algorithm that collects user preference data and applies clustering techniques to extract design features, offering personalised design recommendations. This approach emphasises active consumer participation and preference forecasting, establishing a practical technical framework for customised product development. The

integration of heterogeneous data evaluation methods has further enhanced the scientific robustness of design decision-making. (Lai et al. 2023) introduced a product appearance evaluation method that fuses subjective feedback with objective physiological indicators. Utilising the Probabilistic Linguistic Term Set with Interval Uncertainty to handle ambiguous information, and the Measurement of Alternatives and Ranking according to the Compromise Solution method for comprehensive evaluation, this approach strengthens the reliability of decision-making processes. In summary, the incorporation of advanced algorithmic methods has significantly broadened the capabilities of design tools and provided innovative, data-driven strategies to address dynamic consumer needs and market shifts.

Despite these advances, several limitations remain. Many existing approaches focus on optimising a single design dimension, such as performance or emotional appeal, while lacking mechanisms for multi-objective collaborative optimisation. Additionally, some algorithmic frameworks are highly specific to particular application scenarios, limiting their generalizability across broader design contexts.

To overcome these challenges, the present study proposes an intelligent shape design optimisation method that combines GA with backpropagation (BP) neural networks. By leveraging GA to optimise the weights of BP neural networks, the method balances emotional attributes and design performance in appearance modelling. Morphological analysis is used to deconstruct and encode the components of air purifiers, enabling the generation of diverse design alternatives. Intelligent algorithms are applied to predict emotional responses, offering a flexible and scalable reference framework for the design of various product categories.

Through hierarchical analysis of affective vocabulary and feature coupling analysis, this study constructs a transparent emotional mapping mechanism to guide design decisions. By integrating intelligent algorithms with KE, the research addresses the limitations of traditional single-objective approaches and establishes a user perception oriented optimisation system, offering an efficient and adaptable solution for contemporary product design.

3. Methodology

3.1. Research process

This study proposes an integrated framework for optimising the form design of air purifiers by combining consumer emotional perceptions with intelligent algorithms (Figure 1). The objective is to enhance both market competitiveness and user satisfaction through a systematic analysis of emotional needs and algorithm-driven optimisation. The proposed framework integrates multiple methods, including KE, AHP, CV, CCD, GA, and BP neural networks. These are employed in a structured sequence to enable accurate, data-driven product design optimisation. The framework comprises the following key stages:

- A list of initial affective terms related to the appearance of air purifiers was generated through literature review, expert interviews, and analysis of consumer reviews. This process involved 10 domain experts, including three product designers, two consumer representatives, and five academic researchers specialising in design and psychology.

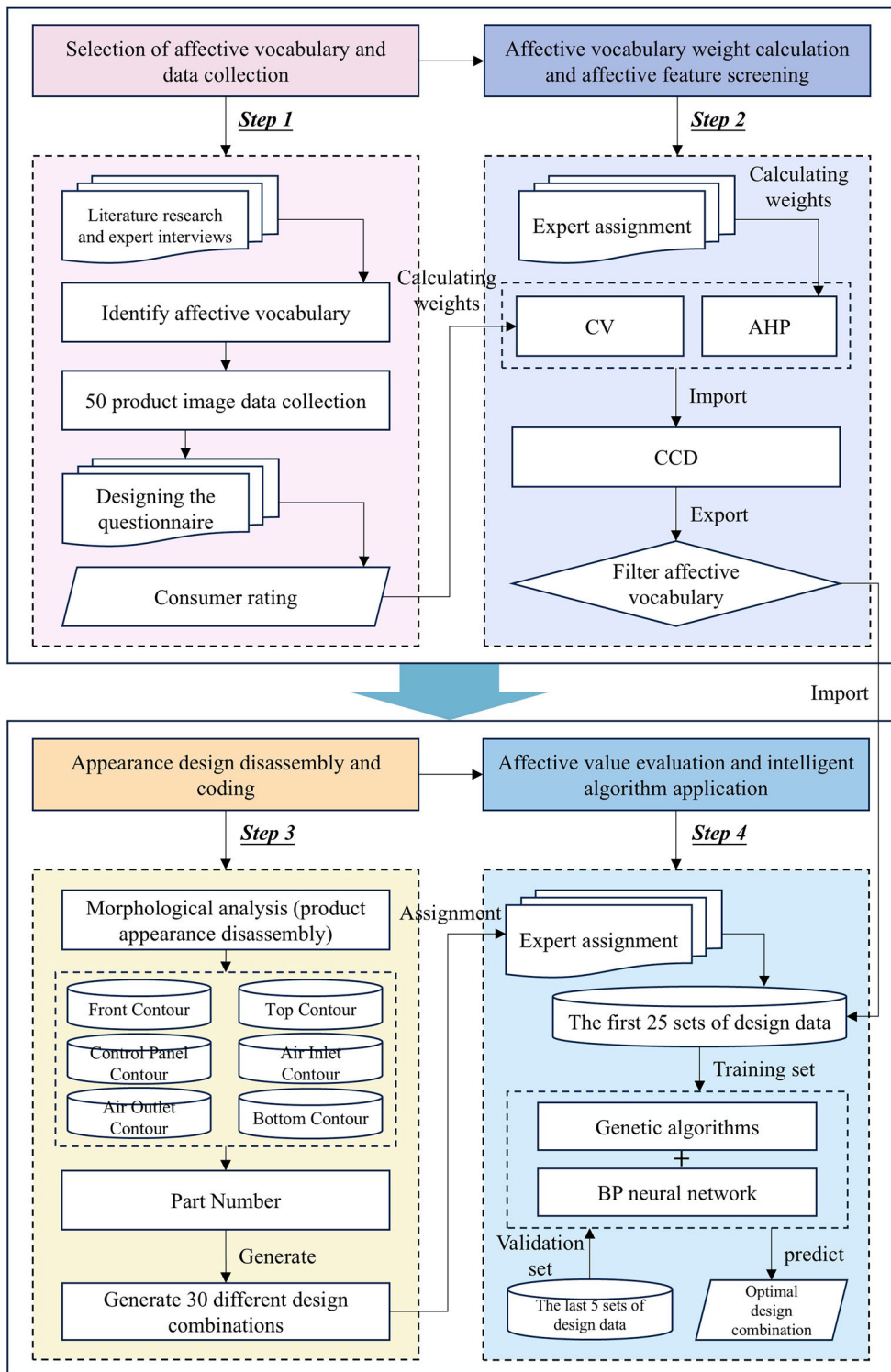


Figure 1. Proposed framework.

Their input ensured that the selected vocabulary was both professionally grounded and emotionally relevant.

- The AHP method was used to calculate the subjective weights of affective vocabulary. A panel of 10 experts (comprising eight professors in industrial design and two senior product design engineers) conducted pairwise comparisons to evaluate the relative importance of each term.
- In parallel, the CV method was applied to derive objective weights based on perceptual evaluations. A total of 20 university-based consumers participated in rating 50 best-selling air purifier models using a 9-point Likert scale. The variability in their responses served as the basis for CV calculations, ensuring a consumer-informed perspective in the final weighting.
- The CCD method was employed to examine the interrelationships among affective terms. This helped identify a subset of core affective descriptors that are both subjectively important and objectively consistent across participants.
- The appearance of air purifiers was decomposed into key morphological elements, which were encoded to allow for modular manipulation and variation in the design phase. This encoding facilitated algorithmic generation of new appearance schemes.
- Based on the encoded morphological elements, 30 distinct air purifier design schemes were algorithmically generated. The same 20 university-based consumers who participated in the earlier CV evaluation were asked to assess these schemes using the core affective vocabulary. This step produced perceptual scores that were used to train and validate the GA-BPNN predictive model.

3.2. Analytic hierarchy process

The AHP is a widely used multi-criteria decision-making method, particularly effective in addressing complex problems that involve both qualitative judgments and quantitative evaluations (Lin et al. 2008). In this study, AHP is applied to determine the relative importance of affective vocabulary in the appearance design of air purifiers. Expert judgments and hierarchical analysis are employed to quantify the weight of each term in the design process. The AHP procedure consists of the following steps:

(1) Construction of the Judgment Matrix:

An expert panel evaluates the importance of each affective term through pairwise comparisons. These comparisons are based on the perceived contribution of each term to the design of air purifier appearances. A 1–9 scale and its reciprocals are used to indicate relative importance (Yang et al. 2021), following Table 1.

(2) Calculation of the Row Product:

For each row i in the judgment matrix, the product of all entries is calculated as follows:

$$M_i = \prod_{j=1}^m b_{ij} (i = 1, 2, \dots, m) \quad (1)$$

Table 1. Scale values in the judgment matrix.

Scale value	Interpretation
1	Two factors are equally important
3	Factor i is slightly more important than factor j
5	Factor i is significantly more important than factor j
7	Factor i is strongly more important than factor j
9	Factor i is absolutely more important than factor j
2, 4, 6, 8	Intermediate values between adjacent judgments

where b_{ij} is the element in the i -th row and j -th column, and m is the number of affective terms.

(3) Calculation of the Geometric Mean:

The geometric mean of each row is computed to obtain a preliminary weight:

$$a_i = \sqrt[m]{M_i} (i = 1, 2, \dots, m) \quad (2)$$

(4) Normalisation of Weights:

The relative weights of the affective terms are derived by normalising the geometric means:

$$w_i = \frac{a_i}{\sum_{i=1}^m a_i} \quad (3)$$

(5) Computation of the Maximum Eigenvalue:

To assess the consistency of the matrix, the maximum eigenvalue λ_{max} is calculated using the following formula:

$$\lambda_{max} = \frac{1}{n} \sum_{i=1}^n \frac{B_{w_i}}{w_i} \quad (4)$$

where B_{w_i} is the i -th component of the product of the judgment matrix and the weight vector, and n is the number of terms.

(6) Consistency Index:

The consistency index is computed to evaluate the deviation from consistency:

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (5)$$

(7) Consistency Ratio:

Table 2. Random consistency index (*RI*).

Matrix order (<i>n</i>)	1	2	3	4	5	6	7	8	9	10
<i>RI</i>	0	0	0.52	0.89	1.12	1.26	1.36	1.41	1.46	1.49

Finally, the consistency ratio is calculated to assess the acceptability of the matrix consistency:

$$CR = \frac{CI}{RI} \quad (6)$$

where *RI* is the random consistency index, determined by the matrix order *n* as shown in Table 2. If $CR \leq 0.1$, the judgment matrix is considered consistent. Otherwise, the matrix must be reevaluated and adjusted.

3.3. Coefficient of variation

The CV is an objective weighting method commonly used in multi-criteria decision-making and product design. It evaluates the importance of different factors by measuring data dispersion, factors with higher variability are typically considered to exert a greater influence on the system (Jalilibal et al. 2021). In this study, the CV method is applied to calculate the objective weights of affective vocabulary in the appearance design of air purifiers. The CV-based weighting process consists of the following steps:

(1) Construction of the Evaluation Matrix:

An evaluation matrix is established where each element x_{ij} represents the score of the *i*-th product sample on the *j*-th affective vocabulary. If there are *n* product samples and *m* affective terms, the matrix is represented as:

$$X_{ij} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1j} \\ x_{21} & x_{22} & \cdots & x_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i1} & x_{i2} & \cdots & x_{ij} \end{bmatrix} \quad (7)$$

Where: $i = 1, 2, 3, \dots, n; j = 1, 2, 3, \dots, m$.

(2) Calculation of the Coefficient of Variation:

For each affective term (column *j*), the mean score and standard deviation are calculated as follows:

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij} \quad (8)$$

where: $\bar{x}_j$ is the mean score for the *j*-th affective vocabulary, *n* is the number of samples, x_{ij} is the score of the *i*-th sample for the *j*-th affective vocabulary.

$$d_j = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} \quad (9)$$

where: d_j is the standard deviation for the j -th affective vocabulary, indicating the dispersion of the scores across all samples.

$$CV_j = \frac{d_j}{\bar{x}_j} \quad (10)$$

where: CV_j is the coefficient of variation for the j -th affective vocabulary.

(3) Normalisation and Weight Calculation:

The final weight of each affective vocabulary term is derived by normalising its CV value:

$$\omega' = \frac{CV_j}{\sum_{j=1}^n CV_j} \quad (11)$$

These normalised weights reflect the relative objective importance of each affective term in the overall product form design.

3.4. Coupling coordination degree method

The CCD method is an analytical tool used to evaluate the interaction and coordination between multiple subsystems and is widely applied in multi-factor evaluation studies (Ren and Qu 2024). In this study, the CCD method is employed to identify the most relevant affective vocabulary for air purifier appearance design by examining the consistency between subjective and objective weights. A higher degree of coordination suggests a better alignment between expert judgment and consumer perception (Dong and Li 2021), indicating that such affective terms more effectively capture user emotional needs and are more suitable for guiding form optimisation. The CCD method in this study involves the following steps:

(1) Calculation of Coupling Degree:

Derived from physics, the coupling degree measures the interdependence between systems. In this context, the subjective weights (from AHP) and objective weights (from CV) are regarded as two systems. The general form of the coupling degree for n systems is:

12) where: $u_1, u_2, \dots, u_n$ are the weights from different systems.

Given that this study involves two systems (subjective and objective weights), the formula simplifies to:

$$C_2 = \left\{ \frac{(u_1 \times u_2)}{(u_1 + u_2) \times (u_1 + u_2)} \right\}^{1/2} \quad (13)$$

where: u_1 and u_2 denote the subjective and objective weights, respectively.

(2) Construction of the Coupling Coordination Degree Model:

After obtaining the coupling degree, the next step is to assess how well the two systems align in terms of contribution to the design objective. This is achieved through a coordination model:

$$D = \sqrt{C \cdot T} \quad (14)$$

Where: C is the coupling degree and T is the coordination index.

Since both subjective and objective weights are considered equally important in this study, the coordination index T is defined as:

$$T = \alpha u_1 + \beta u_2 \quad (15)$$

where: $\alpha = \beta = 0.5$, ensuring an equal-weighted contribution of both systems.

3.5. GA-BP model

The Backpropagation Neural Network (BPNN) is a widely used supervised learning algorithm that iteratively adjusts the weights and thresholds of the network to minimise the difference between predicted outputs and actual target values (Chen and Cheng 2022). In this study, the BPNN is utilised to predict the emotional value of air purifier appearance design schemes, aiming to optimise the design to better reflect consumer emotional preferences. The BPNN consists of three primary layers: an input layer, a hidden layer, and an output layer (Jing et al. 2023):

- Input layer $X = (X_1, \dots, X_n)$: Each input node represents a design feature.
- Hidden layer $h = (h_1, \dots, h_j)$: Neurons in this layer compute the weighted sum of inputs, followed by a nonlinear activation function.
- Output layer $y = (y_1, \dots, y_k)$: This layer produces the predicted emotional values.

The BPNN updates the weights and thresholds using the backpropagation algorithm. The accuracy of the model depends on the weights $W = (W_{ij})$ and thresholds across both the hidden and output layers. A schematic of the network is illustrated in Figure 2.

The GA is a global optimisation technique inspired by the principles of natural selection and genetic inheritance. It is employed in this study to optimise the initial weights and thresholds of the BPNN, enhancing training efficiency and improving prediction accuracy (Ding et al. 2020). The integration of GA with BPNN forms the GA-BP model, which is implemented through the following steps:

(1) Population Initialisation:

Each individual in the GA population encodes a potential solution, representing a complete set of BPNN weights and thresholds. Chromosomes are encoded using real numbers, covering the weights between input and hidden layers, the thresholds of the hidden layer, the weights between hidden and output layers, and the output layer threshold.

(2) Fitness Evaluation:

Fitness is measured using the inverse of the Mean Squared Error (MSE), defined as:

$$F(\mathbf{v}) = \frac{1}{1 + \text{MSE}(\mathbf{v})}, \text{ while } \text{MSE} = \frac{1}{m} \sum_{i=1}^m (\mathbf{y}_i - \hat{\mathbf{y}}_i)^2 \quad (16)$$

Where: m is the number of training samples, $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ are the actual and predicted values for the i -th output node.

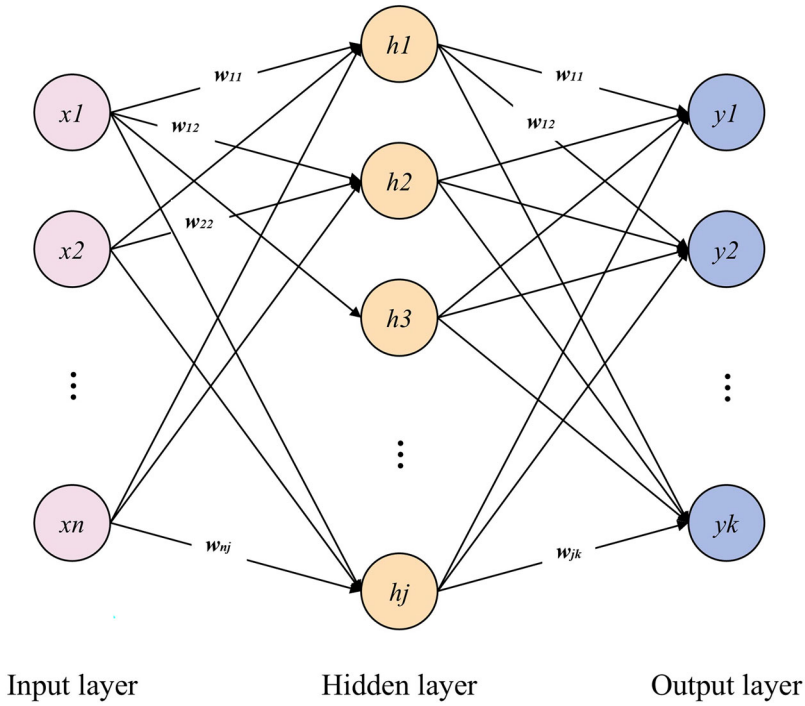


Figure 2. Schematic of BP neural network.

(3) Selection Operation:

The selection mechanism is based on fitness-proportionate (roulette wheel) selection. The probability of selecting individual i is:

$$(P_i) = \frac{F(\mathbf{v}_i)}{\sum_{j=1}^N F(\mathbf{v}_j)} \quad (17)$$

(4) Crossover Operation:

Crossover is used to recombine chromosomes of selected parent individuals to generate offspring. In this study, real-valued crossover is applied, where crossover points are randomly chosen and genes are exchanged between parent chromosomes.

(5) Mutation Operation:

Mutation introduces random changes to selected genes in the chromosomes, maintaining genetic diversity and helping avoid premature convergence.

(6) Determination of Hidden Layer Node Count:

The number of nodes in the hidden layer is a critical hyperparameter. Too many nodes may lead to overfitting, while too few may result in underfitting. An empirical formula is used to estimate the optimal number:

$$j = \sqrt{n + k} + c \quad (18)$$

where: j , n , and k denote the numbers of nodes in the hidden, input, and output layers, respectively, and $c \in [0, 10]$ is an adjustment constant.

(7) Optimisation of Weights and Thresholds:

During training, the output of each layer is computed as follows:

$$\begin{cases} A_1 = (x * w + b_1) * \text{tansig} \\ A_2 = (A_1 * W + b_2) * \text{purelin} \end{cases} \quad (19)$$

where: A_1 and A_2 represent the outputs of the hidden and output layers, x is the input data, w , W , b_1 , and b_2 are the weights and thresholds at each layer, *tansig* and *purelin* are the activation functions of the nodes.

4. Case study

4.1. Identification of affective vocabulary

Air purifiers were chosen as the case study in this research because they are increasingly integrated into everyday home environments and are strongly associated with perceptions of cleanliness, safety, and modern lifestyle. As a result, their appearance is not only functional but also closely tied to users' emotional responses and aesthetic preferences, making them ideal for investigating affective design strategies.

To build a comprehensive affective vocabulary relevant to air purifier design, a three-stage data collection process was conducted:

- Academic journals and design-related publications were reviewed to extract terminology commonly used to describe emotional and aesthetic attributes in product design.
- Semi-structured interviews were conducted with five industrial designers who have experience in home appliance design. Participants were asked to describe the emotional impressions and stylistic trends they associate with air purifiers.
- A sample of 50 commercially available air purifiers from established brands was analysed across e-commerce platforms and design websites. Descriptive words used in marketing texts, customer reviews, and product specifications were extracted and recorded.

Each term collected was recorded on a card along with its source (literature, interview, or market analysis) to maintain traceability. In total, over 20 affective descriptors were gathered at this stage. Following this, a focus group discussion was conducted involving ten participants, including three product designers, two consumer representatives, and five academic researchers in design and psychology. Using the principles of the affinity diagram method (Xu and Song 2024), participants worked collaboratively to sort the cards into

Table 3. Classification of affective vocabulary.

ID	Core affective vocabulary	Descriptive similarity terms
E ₁	Modern	Innovative (E ₁₁), Urban Style (E ₁₂)
E ₂	Minimalist	Intuitive (E ₂₁), Unadorned (E ₂₂)
E ₃	Fashionable	Trendy (E ₃₁), Youthful (E ₃₂)
E ₄	Technological	Intelligent (E ₄₁), Digital (E ₄₂), High-Tech (E ₄₃)
E ₅	High-End	Luxurious (E ₅₁), Exquisite Craftsmanship (E ₅₂)
E ₆	Warm	Intimate (E ₆₁), Homely (E ₆₂), Safe (E ₆₃)
E ₇	Elegant	Delicate (E ₇₁), Graceful (E ₇₂), Balanced (E ₇₃)
E ₈	Natural	Harmonious (E ₈₁), Ecological (E ₈₂), Environmentally Friendly (E ₈₃)

Table 4. Consistency test results.

	Goal level	Criterion level							
		E ₁	E ₂	E ₃	E ₄	E ₅	E ₆	E ₇	E ₈
λ_{max}	8.8072	2.0000	2.0000	2.0000	3.0385	2.0000	3.0385	3.0385	3.0385
CR	0.0818	0.0000	0.0000	0.0000	0.0370	0.0000	0.0370	0.0370	0.0370

thematic clusters based on perceived emotional similarity and contextual relevance. The grouping was done iteratively, with active discussion to clarify interpretations and eliminate ambiguity.

Through this process, eight core affective vocabulary terms were identified, each representing a dominant emotional attribute in air purifier appearance design. Each core term was supplemented with several descriptive similarity terms that clarify its meaning and range. These eight terms form the foundation for the subsequent evaluation and design optimisation processes. The classification results are shown in Table 3.

4.2. Subjective weight analysis

To determine the relative importance of the affective vocabulary terms in shaping consumer emotional preferences for air purifier appearance design, the AHP was employed to assess their subjective weights. A panel of ten experts, including eight professors specialising in industrial design and two senior product design engineers, was invited to perform pairwise comparisons of the eight selected affective descriptors, resulting in a structured judgment matrix.

This matrix was refined using the maximum improvement direction algorithm, which enhances accuracy in expert evaluations (Shao et al. 2023). The resulting weights were calculated and validated through a consistency check, following the procedures outlined in Equations (1) through (6). As shown in Tables 4 and 5, all consistency ratios (CR) are below the standard threshold of 0.1, indicating satisfactory consistency in expert judgments.

4.3. Objective weight analysis

The CV method was used to objectively evaluate the significance of each affective vocabulary term in shaping consumer perceptions, based on the variability in consumer ratings (Zhang et al. 2024). This approach provides a data-driven basis for assigning weights to affective attributes in product design. To carry out the analysis, data were collected from the top 50 best-selling air purifier models on the Tmall online shopping platform (see

Table 5. Weights of affective vocabulary.

Goal Level	Criterion level	Weight	Sub-criterion level	Weight	Composite weight
Affective Vocabulary	E ₁	0.3755	E ₁₁	0.7500	0.2816
			E ₁₂	0.2500	0.1557
			E ₂₁	0.7500	0.0939
	E ₂	0.2075	E ₂₂	0.2500	0.0868
			E ₃₁	0.7500	0.0863
			E ₃₂	0.2500	0.0519
	E ₃	0.1150	E ₄₁	0.6370	0.0465
			E ₄₂	0.2583	0.0352
			E ₄₃	0.1047	0.0344
	E ₄	0.1363	E ₅₁	0.7500	0.0288
			E ₅₂	0.2500	0.0187
			E ₆₁	0.6370	0.0155
	E ₅	0.0620	E ₆₂	0.2583	0.0143
			E ₆₃	0.1047	0.0139
			E ₇₁	0.6370	0.0130
	E ₆	0.0539	E ₇₂	0.2583	0.0076
			E ₇₃	0.1047	0.0056
			E ₈₁	0.6370	0.0053
	E ₇	0.0294	E ₈₂	0.2583	0.0031
			E ₈₃	0.1047	0.0021
	E ₈	0.0205			



Figure 3. Air purifier sample selection.

Figure 3). These samples reflect products with strong market performance and consumer engagement.

Next, a total of 20 consumers were invited to assess the aesthetic design of these 50 models. Each participant rated the degree to which each product embodied each affective vocabulary term, using a 9-point Likert scale (1 = ‘completely inconsistent’; 9 =

Table 6. Consumer evaluation matrix and objective weights.

Affective vocabulary	Sample 1	Sample 2	Sample 3	...	Sample 49	Sample 50	CV	Weight
E ₁	5.89	6.43	7.21	...	7.12	6.34	0.1264	0.1469
E ₂	7.23	6.45	8.12	...	7.65	6.89	0.1263	0.1467
E ₃	6.34	7.1	7.89	...	7.34	6.67	0.1085	0.1261
E ₄	6.78	7.21	8.34	...	8.23	7.34	0.1167	0.1356
E ₅	7.9	6.98	8.56	...	8.78	7.65	0.1040	0.1209
E ₆	7.45	6.89	8.76	...	8.45	7.34	0.1066	0.1239
E ₇	8.56	7.89	9	...	8.56	7.78	0.0887	0.1031
E ₈	8.12	7.65	9	...	8.9	8	0.0834	0.0969

Table 7. Coupling coordination degree classification.

D Value range	Coordination level	Degree of coupling coordination	D value range	Coordination level	Degree of coupling coordination
0.0 ~ 0.1	1	Extremely Uncoordinated	0.5 ~ 0.6	6	Barely Coordinated
0.1 ~ 0.2	2	Severely Uncoordinated	0.6 ~ 0.7	7	Minimally Coordinated
0.2 ~ 0.3	3	Moderately Uncoordinated	0.7 ~ 0.8	8	Moderately Coordinated
0.3 ~ 0.4	4	Slightly Uncoordinated	0.8 ~ 0.9	9	Well-Coordinated
0.4 ~ 0.5	5	Near Uncoordinated	0.9 ~ 1.0	10	Highly Coordinated

'completely consistent'). Mean evaluation scores for each term were calculated across all samples, and objective weights were derived using Equations (7) through (11). The results are presented in Table 6. The CV values indicate the relative dispersion in consumer perception for each affective term, which directly informs their corresponding weights.

4.4. Affective vocabulary selection and coupling coordination analysis

By integrating the results of subjective and objective weight analyses, the CCD method was employed to evaluate the alignment between affective vocabulary and consumer emotional needs. This method involves calculating the coupling degree (C), the coordination index (T), and the coupling coordination degree (D), which together provide a quantitative assessment of consistency between subjective impressions and objective data in product aesthetics (Zou et al. 2022).

The D value serves as a comprehensive indicator for identifying how well each affective vocabulary term corresponds with consumer expectations. A classification system, ranging from 'extremely uncoordinated' to 'highly coordinated', was adopted to interpret D values and guide design decision-making (Dong and Li 2021), as shown in Table 7.

Based on the computed values using Equations (12) through (15), four affective attributes, Modern, Minimalist, Fashionable, and Technological, exhibited the highest degree of coordination. These terms were thus selected as the core affective vocabulary for optimising air purifier design (see Table 8). Their selection provides a robust theoretical foundation for the subsequent design refinement process.

4.5. Product component encoding and morphological analysis

To systematically analyse the morphological design features of air purifiers, this study applies morphological analysis to deconstruct representative product samples and extract their key components and structural characteristics. Based on a review of top-selling

Table 8. Coupling coordination degree of affective vocabulary.

Affective vocabulary	Subjective weight	Objective weight	C	T	D	Coordination level
E ₁	0.3755	0.1469	1.000	0.990	0.995	10
E ₂	0.2075	0.1467	0.953	0.756	0.849	9
E ₃	0.1150	0.1261	0.931	0.427	0.630	6
E ₄	0.1363	0.1356	0.917	0.549	0.709	8
E ₅	0.0620	0.1209	0.809	0.302	0.495	5
E ₆	0.0539	0.1239	0.732	0.321	0.485	5
E ₇	0.0294	0.1031	0.812	0.083	0.260	3
E ₈	0.0205	0.0969	1.000	0.010	0.100	2

Table 9. Component encoding of air purifier morphologies.

Design Element	Code 1	Code 2	Code 3	Code 4	Code 5
Top Contour (A)					
Bottom Contour (B)					
Front Contour (C)					
Control Panel Contour (D)					
Air Inlet Contour (E)					
Air Outlet Contour (F)					

market models, six critical design elements were identified: top contour (A), bottom contour (B), front contour (C), control panel contour (D), air inlet contour (E), and air outlet contour (F).

Each of these elements was further classified into morphological subcategories and encoded accordingly, as summarised in Table 9. In the proposed encoding system, primary components are represented by letters A-F, while their specific morphological variations are indicated by numerical values ranging from 1 to 5. This approach not only standardises the morphological description of air purifier designs but also generates a structured dataset that can be used as input for subsequent neural network training and design optimisation tasks.

4.6. Determination of input layer indicators

Based on Table 10, thirty representative air purifier samples were selected, and their primary design elements were encoded numerically. These design elements, including top contour (A), bottom contour (B), front contour (C), control panel contour (D), air inlet contour (E), and

Table 10. Design element encoding and normalised affective evaluation values.

Sample	Design element encoding						Core affective vocabulary evaluation			
	A	B	C	D	E	F	Modern	Minimalist	Fashionable	Technological
1	0.667	0.000	0.250	0.000	0.000	0.000	0.357	0.194	0.473	0.354
2	1.000	1.000	0.250	0.667	0.500	0.000	0.000	0.000	0.000	0.000
3	0.667	0.000	0.250	0.333	0.000	1.000	0.086	0.156	0.090	0.199
4	0.333	0.500	0.000	0.000	0.000	1.000	0.793	0.810	0.785	0.841
5	0.667	1.000	0.250	0.000	1.000	1.000	0.046	0.031	0.043	0.079
6	0.667	0.000	0.500	0.000	0.500	0.500	0.604	0.696	0.566	0.682
7	1.000	1.000	0.250	0.333	1.000	0.000	0.286	0.308	0.262	0.361
8	0.333	0.500	0.000	0.000	0.500	1.000	0.882	0.848	0.871	0.964
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
28	0.667	0.000	0.250	0.333	0.500	0.000	0.196	0.118	0.211	0.199
29	1.000	1.000	0.250	0.333	0.000	0.000	0.721	0.654	0.688	0.762
30	0.667	0.000	0.250	0.333	0.000	0.500	0.918	0.886	0.914	0.755

air outlet contour (F), served as the input indicators for the neural network model. Accordingly, the input layer was configured with six neurons, corresponding to the six encoded design variables.

Simultaneously, the normalised evaluation scores of four core affective vocabulary dimensions, Modern, Minimalist, Fashionable, and Technological, were used as output indicators, resulting in a neural network output layer with four neurons.

To ensure comparability between input and output data, min–max normalisation was applied to all input variables, as shown in Table 10. This preprocessing step mapped the original values onto a [0,1] interval, thereby minimising the influence of magnitude differences across indicators. Such normalisation is essential for reducing training errors and enhancing the accuracy and stability of the neural network model.

4.7. Construction and training of the GA-BP model

A hybrid GA-BP neural network (see Table 11 for model parameters) was developed using the PyTorch framework to model the relationship between morphological features and affective perception. The network topology consists of three layers: a 6-neuron input layer (representing the six encoded design elements), a hidden layer with 10 neurons activated by the ReLU function, and a 4-neuron output layer corresponding to normalised scores of four core affective vocabulary (Modern, Minimalist, Fashionable, Technological), using a linear activation function. The number of hidden units was determined empirically ($c = 4$) to balance model complexity and generalisation.

The GA encodes network parameters into real-valued vectors. Each individual represents 114 parameters in total: $\mathbf{v} = [\mathbf{W}_h, \mathbf{b}_h, \mathbf{W}_o, \mathbf{b}_o] \in \mathbb{R}^{114}$, where $\mathbf{W}_h \in \mathbb{R}^{6 \times 10}$ and $\mathbf{W}_o \in \mathbb{R}^{10 \times 4}$ are weight matrices for the hidden and output layers, and $\mathbf{b}_h \in \mathbb{R}^{10}$ and $\mathbf{b}_o \in \mathbb{R}^4$ are bias vectors..

The GA optimisation process begins with the initialisation of $N = 20$ individuals, whose parameters are uniformly sampled from the range $\mathcal{U}(-1, 1)$. Fitness is evaluated based on Mean Squared Error (MSE) across $m = 25$ training samples. Selection is performed using the roulette wheel strategy to retain high-fitness individuals while preserving diversity.

Table 11. GA-BP model parameters.

Parameter Name	Value
Population Size	20
Number of Generations	50
Crossover Probability	0.8
Mutation Probability	0.1
Individual Initialisation Range	$[-1, 1]$
Mutation Perturbation Magnitude	$[-0.1, 0.1]$
Input Layer Size	6
Hidden Layer Size	10
Output Layer Size	4
Hidden Layer Activation Function	ReLU
Output Layer Activation Function	Linear

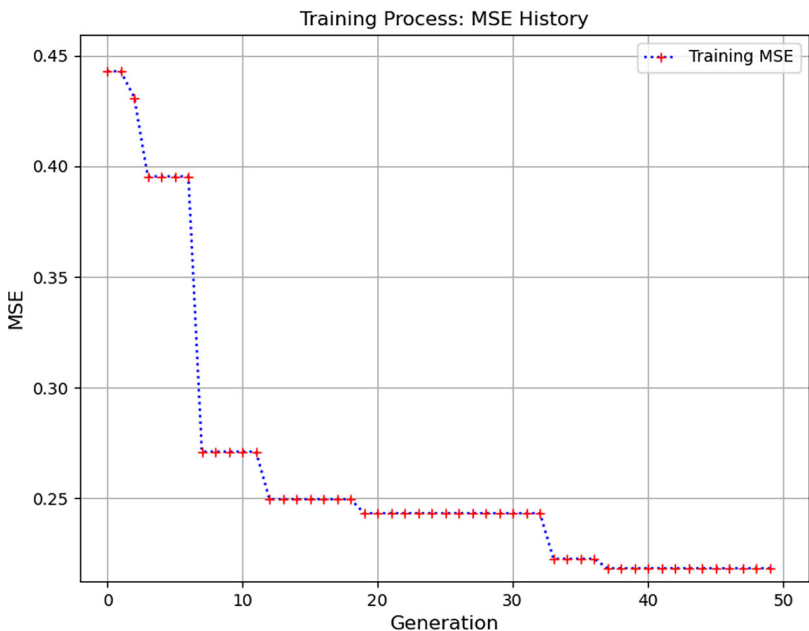


Figure 4. MSE value during training.

Crossover and mutation are then applied. Single-point crossover is performed with a probability of $p_c = 0.8$, randomly exchanging gene segments between parents. Gaussian mutation occurs with a probability of $p_m = 0.1$, introducing perturbations drawn from $\Delta \mathbf{v} \sim \mathcal{N}(0, 0.1)$ to prevent premature convergence.

The algorithm runs for up to $T = 50$ generations or until the best fitness remains unchanged within a threshold of 10^{-4} for 10 consecutive generations. The model was implemented using PyTorch 1.12.1, with GA operations customised. All training was conducted on an NVIDIA RTX 3090 GPU, with each iteration completing in under 20 s (Figure 4).

4.8. Design practice

In this study, the external form of air purifiers was deconstructed into six fundamental components, each further divided into multiple morphological factors. Based on the predictive



Figure 5. Design scheme renderings.

results of the GA-BP neural network model for user affective evaluation, a series of design schemes were generated. Three-dimensional renderings of these schemes are shown in Figure 5. By systematically combining different morphological factors, a total of 2880 unique design configurations were derived ($4 \times 3 \times 5 \times 4 \times 3 \times 4 = 2880$).

All encoded design configurations were input into the GA-BP model for affective prediction. The combinations of morphological elements that yielded the highest predicted values for the four key affective descriptors, modern, minimalistic, stylish, and technological, were identified as follows:

- Modern: A1, B3, C2, D2, E2, F4 (predicted value = 0.904).
- Minimalistic: A2, B2, C3, D1, E3, F2 (predicted value = 0.873).
- Stylish: A4, B3, C5, D3, E1, F4 (predicted value = 0.951).
- Technological: A2, B2, C2, D4, E1, F1 (predicted value = 0.892).

By integrating GA with a BPNN, the GA-BP model effectively overcomes the limitation of local optima in high-dimensional parameter spaces, thereby enhancing global optimisation performance. The GA provides robust global search capabilities, while the BPNN captures complex nonlinear relationships through multi-level mappings. This method significantly improves the accuracy of affective evaluation predictions in air purifier form design, demonstrating strong potential for application in product form optimisation.

5. Discussion

5.1. Model performance and validation of design schemes

To assess the predictive accuracy and practical utility of the GA-BP neural network model in air purifier form design, a comparative analysis was conducted between the model's predicted affective scores and actual consumer evaluations. As shown in Table 12, predicted affective values were compared with the average user ratings, and relative errors

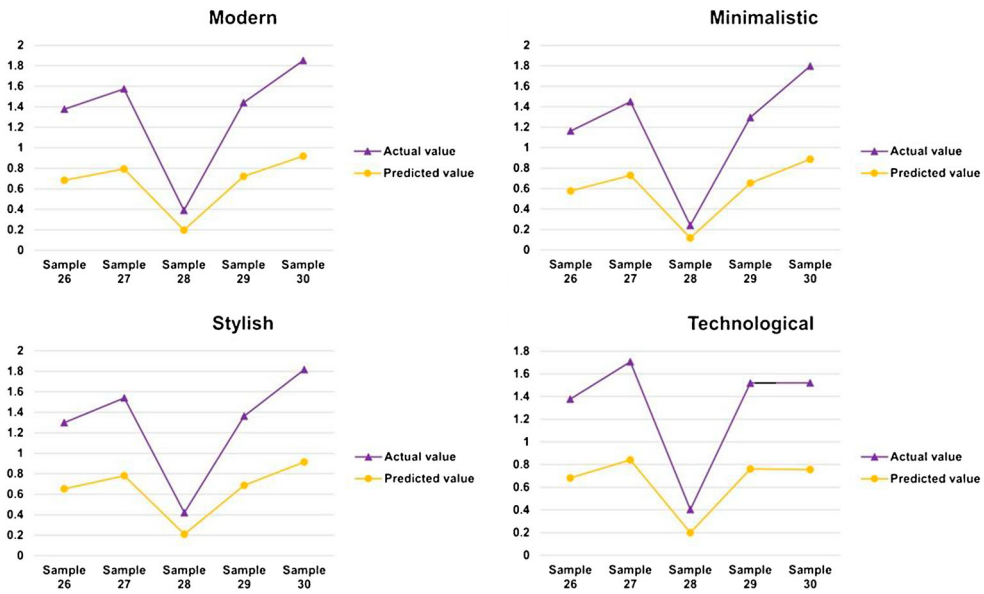


Figure 6. Model performance validation.

Table 12. Comparison of predicted and actual affective values.

Key affective vocabulary	Data error	Sample 26	Sample 27	Sample 28	Sample 29	Sample 30
Modern	Predicted Value	0.682	0.793	0.196	0.721	0.918
	Actual Value	0.695	0.781	0.193	0.718	0.933
	Relative Error (%)	1.91	1.54	1.55	0.42	1.63
Minimalistic	Predicted Value	0.578	0.730	0.118	0.654	0.886
	Actual Value	0.584	0.719	0.121	0.641	0.912
	Relative Error (%)	1.04	1.53	2.54	2.03	2.93
Stylish	Predicted Value	0.652	0.781	0.211	0.688	0.914
	Actual Value	0.647	0.759	0.209	0.674	0.902
	Relative Error (%)	0.77	2.9	0.96	2.08	1.33
Technological	Predicted Value	0.682	0.841	0.199	0.762	0.755
	Actual Value	0.694	0.864	0.204	0.757	0.765
	Relative Error (%)	1.76	2.74	2.51	0.66	1.32

were calculated accordingly. The results presented in Figure 6 demonstrate high predictive accuracy, with relative errors for all core affective descriptors remaining below 3%. These findings validate the effectiveness and robustness of the GA-BP model in optimising product aesthetics.

Specifically, Samples 26–30 were selected from the design solution set for empirical validation. Each sample corresponds to a unique morphological structure derived from the morphological analysis process. Their specific form encodings are as follows:

- Sample 26: (A3, B3, C3, D1, E2, F2).
- Sample 27: (A4, B3, C3, D2, E2, F2).
- Sample 28: (A3, B1, C2, D2, E2, F2).
- Sample 29: (A4, B3, C2, D2, E1, F2).
- Sample 30: (A3, B1, C2, D2, E1, F3).

Table 13. Comparison of weighting methods.

Method combination	Weight values and coordination levels of affective vocabulary							
	E ₁	E ₂	E ₃	E ₄	E ₅	E ₆	E ₇	E ₈
AHP	0.3755	0.2075	0.115	0.1363	0.062	0.0539	0.0294	0.0205
CV	0.1469	0.1467	0.1261	0.1356	0.1209	0.1239	0.1031	0.0969
Coordination Level	10	9	7	8	5	5	3	2
PC	0.2344	0.2031	0.1406	0.1719	0.1094	0.0781	0.0469	0.0156
EW	0.1506	0.1722	0.1440	0.1327	0.1071	0.1063	0.1008	0.0863
Coordination Level	10	10	8	8	6	6	5	2
DARES	0.7424	0.1061	0.0273	0.0818	0.0273	0.0091	0.0030	0.0030
CRITIC	0.1468	0.1112	0.1069	0.1318	0.1124	0.1031	0.1289	0.1589
Coordination Level	10	4	3	5	3	2	3	4

To further evaluate the reliability of the proposed optimal design solutions, 30 consumers participated in a perception survey using a questionnaire covering four affective descriptors: modern, minimalistic, stylish, and technological. The average scores from consumer evaluations were 0.913, 0.826, 0.964, and 0.854, respectively. The model's predicted values were closely aligned with actual user perceptions, with a margin of error within ± 0.5 .

These results demonstrate a strong alignment between computational predictions and human emotional responses, confirming the practical viability of the GA-BP model in guiding affect-driven product design. This approach enables companies to reduce the costs and inefficiencies associated with trial-and-error development processes, while better aligning product form with user expectations. In doing so, it enhances market competitiveness and reduces design risk. More broadly, this research bridges the gap between designers and users by translating consumer affective needs into tangible product attributes, thereby facilitating the successful development of emotionally resonant, consumer-driven products.

5.2. Comparison of weighting methods

In this study, a hybrid weighting approach combining the AHP and CV methods was employed to identify and optimise the core affective descriptors for air purifier design. Two additional combinations of subjective – objective weighting methods were also examined for comparison:

- Precedence Chart (PC) and Entropy Weight (EW) Methods
- Decision Alternative Ratio Evaluation System (DARES) and Criteria Importance Through Intercriteria Correlation (CRITIC) Method

A coupling coordination degree model was used to assess the coordination levels of affective descriptors under each method, and the comparative results are summarised in Table 13.

(1) Coordination Analysis: PC and EW Methods

The PC method allocates weights based on expert judgment, offering a structured ranking of descriptors (Liu, Lu, and Qiang 2024), whereas the EW method assigns importance

based on the degree of information entropy in the dataset (Ren 2020), capturing the dispersion of consumer responses. However, a key limitation of the EW method is its tendency to distribute weights relatively evenly across descriptors, which can reduce discriminative power.

This characteristic explains why the PC – EW combination resulted in the selection of a broader range of core descriptors (six total). While this approach ensures balanced weight assignment, its lower sensitivity may dilute the focus of design decisions, potentially impacting the precision and distinctiveness of the resulting design outcomes.

(2) Coordination Analysis: DARES and CRITIC Methods

The combination of DARES and CRITIC yielded the lowest coordination levels, especially for descriptors E_2 through E_8 , where coordination scores were below 5. This method failed to effectively filter key descriptors and displayed high weight concentration on E_1 (subjective weight > 0.7), indicating overemphasis.

DARES determines descriptor importance through iterative pairwise comparisons (Xiang, Qin, and Yin 2020), offering adaptability but also posing the risk of extreme weight allocation. CRITIC, which emphasises variability and conflict among criteria (Yan et al. 2022), may underweight descriptors with low variability – leading to incomplete representation of user sentiment.

Overall, the DARES – CRITIC combination demonstrated weak coordination and poor robustness in descriptor selection, making it unsuitable for applications demanding balance and generalizability.

(3) Comparative Summary

Among the three combinations, the AHP – CV method showed the highest coordination and overall stability, making it the most suitable for affective product form design. Its strength lies in the balanced integration of structured expert judgment (AHP) with user-perception variability (CV). AHP ensures logical and experience-based descriptor prioritisation, while CV reflects the diversity of user sentiment across different design schemes.

This balance prevents excessive focus on a single descriptor (as with DARES – CRITIC) or overgeneralisation (as with PC – EW), producing a focused yet flexible weighting framework. In the context of this study, where a GA-optimized BP neural network models the nonlinear mapping between design features and emotional responses, such stability is critical. Inconsistent or biased weights may distort training and reduce prediction reliability.

By selecting a concise and representative set of affective descriptors (E_1 – E_4), the AHP – CV method supports high-accuracy emotional prediction, robust model training, and improved user experience. It enables the extraction of emotionally salient design features and facilitates intelligent, consumer-centered product development.

6. Conclusion

This study proposes an integrated framework that connects user perception with the complex morphological features of industrial products, using air purifiers as a representative case. By combining multi-source affective vocabulary weighting techniques (AHP, CV, and

CCD) with morphological decomposition and a GA-optimized BP neural network, the model enables accurate prediction of emotional responses and supports data-informed form optimisation.

The key contribution lies in embedding the GA-BPNN model within a structured design evaluation pipeline, rather than applying it in isolation. This integration enhances the model's nonlinear mapping capacity, improves convergence stability, and strengthens the scientific validity of emotion-driven product form generation. The study thus contributes both methodologically and practically to the fields of Kansei Engineering and intelligent product design.

Despite these contributions, several limitations should be acknowledged. First, the method involves technical complexity and significant computational demand, which may limit accessibility for non-expert users. Second, while affective vocabulary weighting and prediction were addressed systematically, the identification of critical form features remains a challenge. Third, the emotional evaluation was conducted by only 20 university-based participants, introducing potential perceptual bias and limiting generalizability. Moreover, the focus on air purifiers as the sole product category restricts the broader applicability of the framework. Lastly, the study did not directly compare the GA-BPNN with alternative prediction models (e.g. SVM, Random Forest, or deep learning), nor did it use external datasets for validation.

Future research should pursue broader empirical validation by incorporating more diverse participant groups and product types. Comparative benchmarking against other prediction models, along with cross-domain dataset testing, will further establish the framework's feasibility and robustness. Additionally, efforts should be made to develop user-friendly interfaces and integrate advanced feature selection and dimensionality reduction techniques to improve model interpretability, usability, and computational efficiency. Incorporating user segmentation and personalised affective modelling may also help tailor design strategies to diverse emotional preferences across consumer groups.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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